

Interventions & Events

AE, CM and EX — the general observation classes

DM was one row per subject. From here on, subjects have MANY records — which means sequence numbers, study days, and a lot more timing.

By the end of this module you can...

1

Tell Interventions from Events and know which domains fall into each class.

2

Read the general-observation structure topic, qualifiers and timing — the pattern every non-Special-Purpose domain follows.

3

Explain AETERM vs AEDECOD and why a submission keeps both the verbatim and the coded term.

4

Distinguish severity from seriousness the single most confused pair in the whole standard.

5

Derive --SEQ and --DY the two variables DM never needed.

Done TO the subject, or happened TO the subject?

INTERVENTIONS

Something given to, or done to, the subject — deliberately.

EX Exposure — the study drug itself

CM Concomitant medications — everything else they took

PR Procedures

Topic variable: `--TRT` (what was given)

`EXTRT` · `CMTRT`

EVENTS

Something that happened to the subject — not planned.

AE Adverse events — the safety backbone

MH Medical history — conditions before the study

DS Disposition — milestones and how they left

Topic variable: `--TERM` (what happened)

`AETERM` · `MHTERM` · `DSTERM`

The test: was it deliberate (Intervention) or did it just happen (Event)? Aspirin taken for a headache is an Intervention; the headache is an Event.

THE PATTERN

Every general-observation domain has the same skeleton

1

IDENTIFIERS

Who and which record

STUDYID · DOMAIN · USUBJID · --SEQ

2

TOPIC

What the record is about

AETERM · CMTRT · EXTRT

3

QUALIFIERS

Everything describing the topic

AESEV · AESER · AEREL · AEOUT · AEDECOD

4

TIMING

When it happened

AESTDTC · AEENDTC · AESTDY · AEENDY

This is what DM did NOT have. DM is Special Purpose — no topic variable, no --SEQ. From here on every domain follows the skeleton above, which is why learning it once pays off six times.

Anatomy of an AE record

USUBJID	AESEQ	AETERM	AEDECOD	AESEV	AESER	AESTDTC	AEENDTC	AESTDY
ABC-01-01-001	1	bad headache	Headache	MODERATE	N	2024-03-15	2024-03-16	15

Identifiers

USUBJID + AESEQ uniquely identify this row

Topic

AETERM — what the investigator reported, verbatim

Qualifiers

AEDECOD, AESEV, AESER, AEREL, AEOUT describe it

Timing

AESTDTC / AEENDTC in ISO, AESTDY / AEENDY as study days

One row per event, per subject. A subject with three adverse events gets three rows — AESEQ 1, 2, 3.

TWO TERMS, BOTH REQUIRED

AETERM and AEDECOD

AETERM

VERBATIM — exactly what the investigator wrote.

"bad headache"

Never cleaned, never corrected, never abbreviated. It is the legal record of what was reported.



AEDECOD

CODED — the dictionary's preferred term.

"Headache"

Assigned by trained coders using MedDRA. Lets you count how many subjects had headache across the whole study.

Why keep both? The verbatim preserves what actually happened; the coded term makes it countable. Three investigators writing "bad headache", "headache" and "cephalalgia" all code to one preferred term — and only then can you say how many subjects had it.

⚠ **Coding is not a lookup table.** MedDRA is a licensed dictionary applied by trained coders — you cannot derive AEDECOD with a CASE statement. In this bootcamp the coded values are ILLUSTRATIVE stand-ins so the exercise runs.

Severity is NOT seriousness

AESEV — Severity

How BAD did it feel?

MILD · MODERATE · SEVERE

A clinical judgement of intensity.

AESER — Seriousness

Did it meet a REGULATORY threshold?

Y / N

Death, life-threatening, hospitalisation, disability, congenital anomaly, or medically important.

A mild rash that required hospitalisation

MILD

Y

Serious but not severe

Severe pain, resolved at home

SEVERE

N

Severe but not serious

Worsening hypertension, subject withdrawn

SEVERE

Y

Both — our ABC-01 case

They are independent. Never derive one from the other — a serious event is not automatically severe, and vice versa.

--SEQ: numbering a subject's records

DM never needed this. Now that a subject can have many rows in one domain, every row needs a unique number within that subject.

USUBJID	AESEQ	AETERM	AESTDTC
ABC-01-01-001	1	bad headache	2024-03-15
ABC-01-01-001	2	Nausea	2024-03-20
ABC-01-01-004	1	fatigue	2024-03-18
ABC-01-01-004	2	worsening hypertension	2024-03-22

The rule

Sequential 1, 2, 3... WITHIN each subject. It restarts at 1 for the next subject — it is not a row number for the whole dataset.

The test

USUBJID + AESEQ must be UNIQUE. That pair is what identifies any single observation in the study.

Sort deterministically before you number — otherwise the same program gives different --SEQ values on different runs.

NEW IN THIS MODULE

--DY: study day, and the missing Day 0

The rule

```
if  --DTC >= RFSTDTC :  --DY = (--DTC - RFSTDTC) + 1      <- on/after first dose
if  --DTC < RFSTDTC :  --DY = (--DTC - RFSTDTC)           <- before, stays negative
```

First dose

2024-03-01

Day 1

Headache starts

2024-03-15

Day 15

Screening visit

2024-02-20

Day -10

There is no Day 0. The day before first dose is Day -1. If your code ever produces a 0, the +1 is missing on one branch.

Ongoing events

Two of the ten ABC-01 adverse events had not resolved when the data was collected. Their end date is blank — and it must stay blank.

AEENDTC	Leave NULL	Do not substitute today's date, the last visit, or the study end.
AEENDY	Leave NULL	No end date means no end study day. Deriving one invents data.
AEOUT	NOT RECOVERED/NOT RESOLVED	The outcome carries the information that it is ongoing.

Why this matters. Inventing an end date would make an unresolved event look resolved. In a safety review that is the difference between an ongoing problem and a closed one — which is exactly the kind of thing a regulator is looking for.

PUTTING IT TOGETHER

raw AE → SDTM AE

AETERM

verbatim, unchanged

bad headache → bad headache

AESTDTC

MIXED formats → ISO 8601

15/03/2024 → 2024-03-15

AESER

normalise → N / Y

No, N → N

AESQ

derive per subject

1, 2, 3 ...

AEDECOD

dictionary coded (MedDRA)

bad headache → Headache

AESEV

trim + upper-case → CT

moderate → MODERATE

AEOUT

DECODE the number → CT

1 → RECOVERED/RESOLVED

AESTDY

derive from RFSTDTC

2024-03-15 → 15

The AE date conversion is the hardest in the study: ONE column contains two different formats. You must detect the format per value.

Common AE mistakes



Cleaning up AETERM

The verbatim term is a legal record. Fix nothing — not spelling, not capitalisation.



Deriving AEDECOD in code

MedDRA coding is done by trained coders with a licensed dictionary.



Confusing severity and seriousness

They are independent. A mild event can be serious.



Assuming one date format

AESTDT holds BOTH DD/MM/YYYY and DD-Mon-YYYY. Detect per value.



Filling in a blank end date

Ongoing means ongoing. Null AEENDTC and null AEENDY.



Numbering --SEQ across subjects

AESEQ restarts at 1 for each subject; it is not a row number.

WHAT'S NEXT

Build AE, then the Interventions

1

Notebook 05 • AE Mixed dates, CT, the AEOUT code decode, AESEQ and AESTDY — in SAS.

2

Day 6 • CM and EX Same skeleton, Interventions instead of Events. Much of your AE code transfers.

3

Then Findings VS and LB — where the structure changes shape again.

AE is the hardest domain you will build. Everything after it reuses what you learn here.